

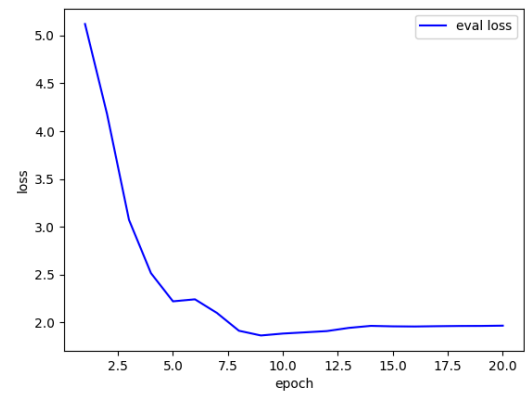
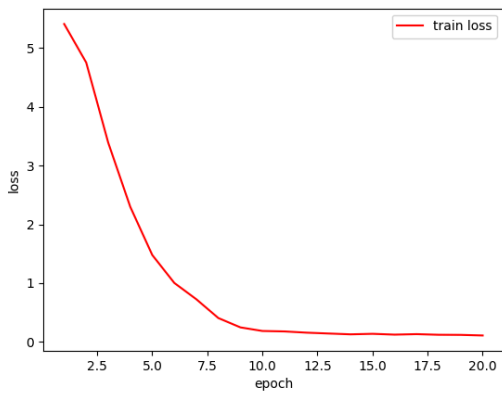
HW7

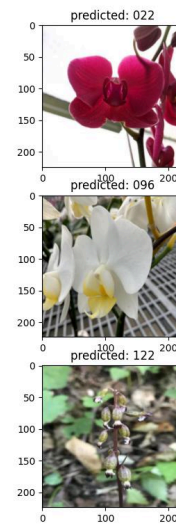
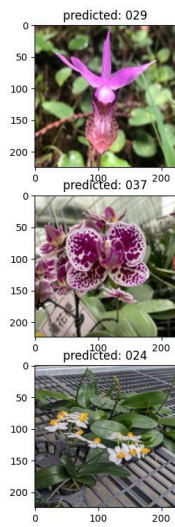
Execution description

https://colab.research.google.com/drive/1LtpFa8IMdUj02QrTpgk1_ngXOBeor95x?usp=sharing

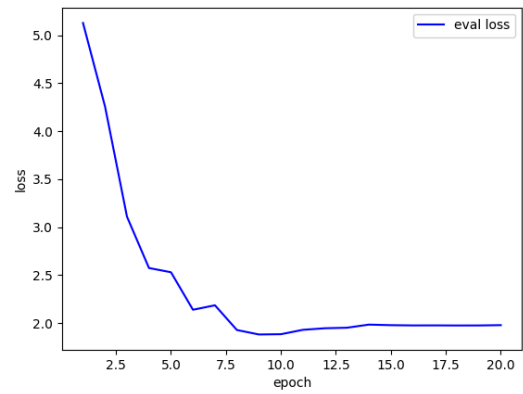
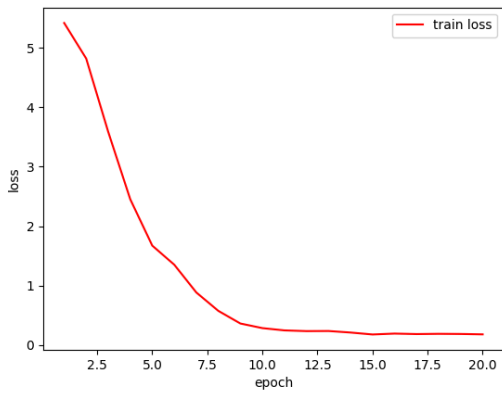
Expermental results

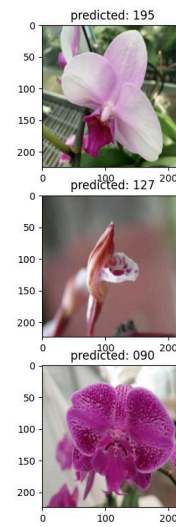
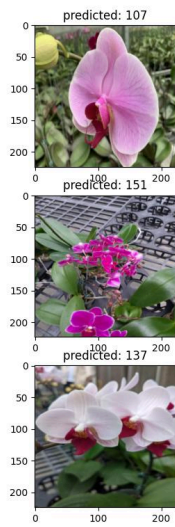
normal



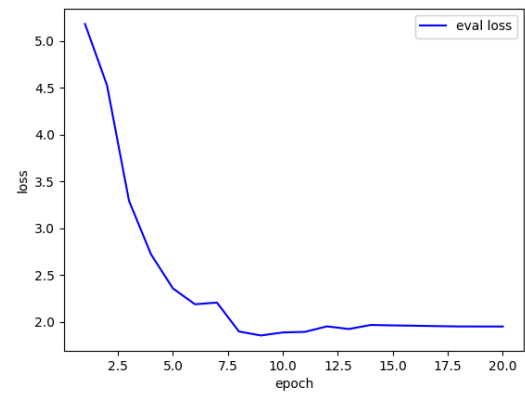
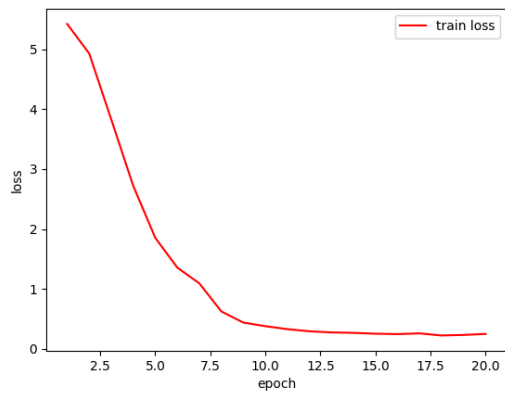


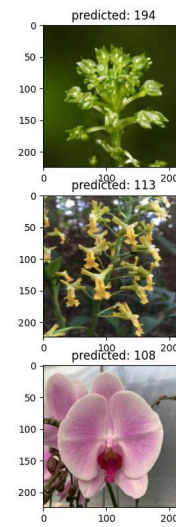
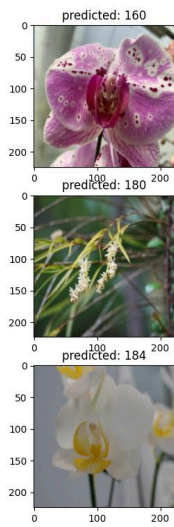
RandomHorizontalFlip





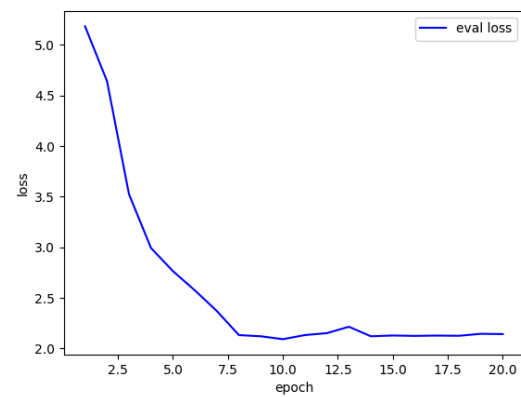
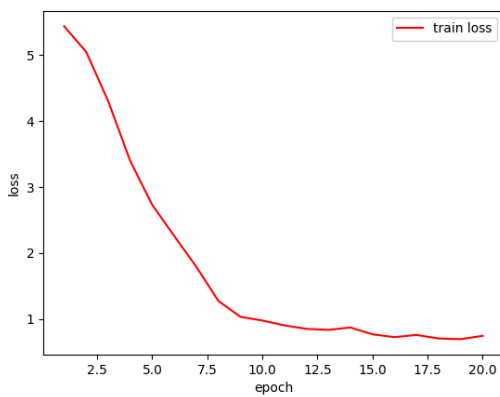
RandomRotation(degrees=15)

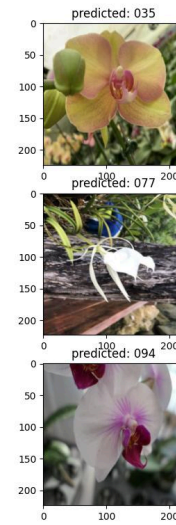
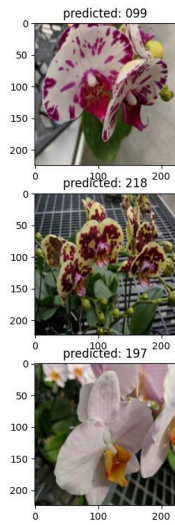




Strong1

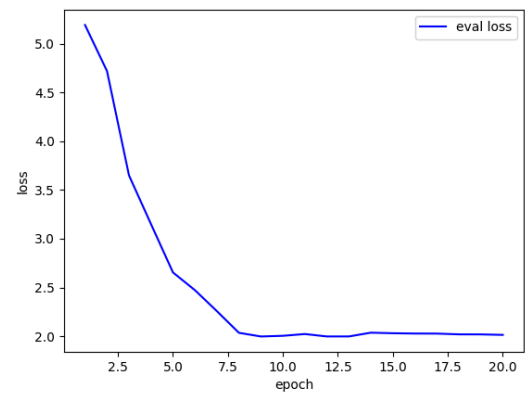
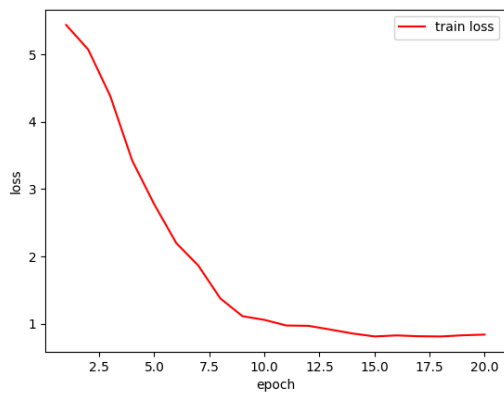
- 隨機水平翻轉影像
- 隨機旋轉影像 ± 15 度
- 隨機調整亮度、對比、飽和度、色調
- 隨機裁剪 80%~100% 區域並縮放到 224×224

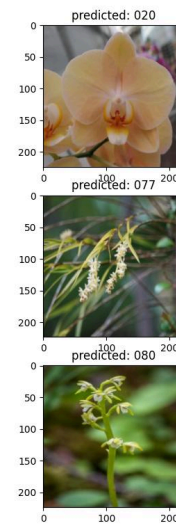
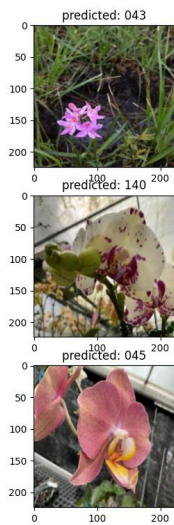




Strong2

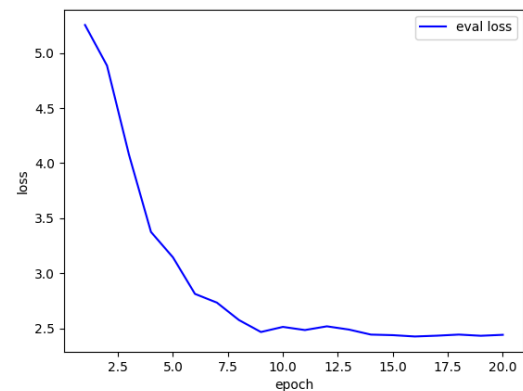
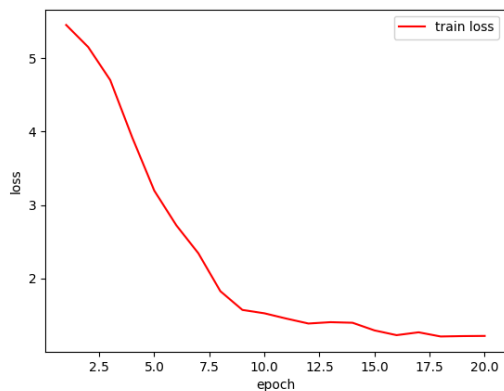
- 隨機水平翻轉影像
- 隨機垂直翻轉影像
- 隨機旋轉影像 ± 30 度
- 隨機套用高斯模糊 (3×3 核心)
- 隨機調整亮度、對比

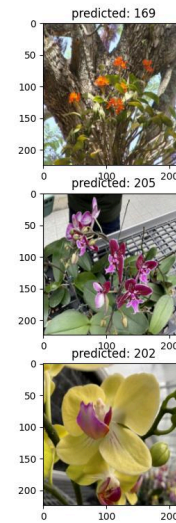
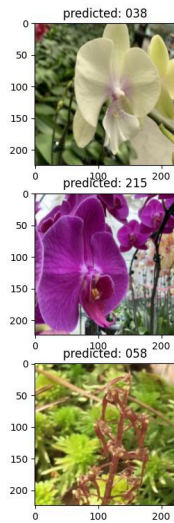




Strong3

- 隨機裁剪 224×224 區域
- 隨機仿射變換（旋轉 ± 20 度、平移 10%、縮放 0.9~1.1 倍）
- 隨機水平翻轉影像
- 隨機調整飽和度、色調
- 以 10% 機率隨機轉為灰階





Conclusion

可以發現, 做了更多的資料增強, 卻反而使模型準確率下降, 觀察後可發現資料集其實很小, 而測試集與訓練集其實蠻相似的, 因此對小資料集或測試集與訓練集極像的情況, 過多或過強的資料增強可能反而害模型學到不必要的變化, 降低測試表現。增強要適度、與任務場景對齊。

Discussion

這次因為規模很小, 運行的速度其實很快, 我也因此學到了並非做越多資料增強就會讓準確率變高, 有時候只會讓模型學到一些不存在的變化